KAIZEN MATHS

Lesson starter

Trigonometric Differentiation

Mixed Trig Challenges

Name _____

S1

Differentiate with respect to x.

$$y = \frac{\sec(-2x+3)}{\cos(-7x-8)}$$

S2

Differentiate with respect to x.

$$y = (\sec(2x-1))^2 \cot(2x-7)$$

S3

Differentiate with respect to x.

$$y = \csc(8x+5)\sin(-9x-2)$$

S4

Differentiate with respect to x.

$$y = \frac{\sec(7x-8)}{\sin(3x-1)}$$

Show your working

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ALG-C0E7-L4 · S

Show your working

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Exit ticket

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E1

Differentiate with respect to x.

$$y = \frac{\sin(-9x - 1)}{\sec(5x + 9)}$$

E2

Differentiate with respect to x.

$$y = (\sin(4x^2 - 5x - 2))^4$$

E3

Differentiate with respect to x.

$$y = \frac{\csc(6x + 5)}{\sin(7x - 2)}$$

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ALG-C0E7-L4 · E

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ALG-C0E7-L4 · E

Show your working

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

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S1

Differentiate with respect to x.

$$y = \frac{\sec(-2x+3)}{\cos(-7x-8)}$$

Answer:

$$\frac{dy}{dx} = \frac{(-2\sec(-2x+3)\tan(-2x+3))\cos(-7x-8) - \sec(-2x+3)(7\sin(-7x-8))}{(\cos(-7x-8))^2}$$

Working:

Identify the numerator and denominator: $u = \sec(-2x+3)$ and $v = \cos(-7x-8)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (\sec(-2x+3)\tan(-2x+3))(-2)$

$u' = -2\sec(-2x+3)\tan(-2x+3)$

$v' = (-\sin(-7x-8))(-7)$

$v' = 7\sin(-7x-8)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(-2\sec(-2x+3)\tan(-2x+3))\cos(-7x-8) - \sec(-2x+3)(7\sin(-7x-8))}{(\cos(-7x-8))^2}$$

S2

Differentiate with respect to x.

$$y = (\sec(2x-1))^2\cot(2x-7)$$

Answer:

$$\frac{dy}{dx} = 4(\sec(2x-1))^1\sec(2x-1)\tan(2x-1)\cot(2x-7) + (\sec(2x-1))^2 - 2\csc^2(2x-7)$$

Working:

This is a product, and the first factor also needs the chain rule.

Let $f = (\sec(2x-1))^2$ and $g = \cot(2x-7)$.

$f' = 2(\sec(2x-1))^{2-1}(\sec(2x-1)\tan(2x-1))(2)$

$f' = 4(\sec(2x-1))^1\sec(2x-1)\tan(2x-1)$

$g' = (-\csc^2(2x-7))(2)$

$g' = -2\csc^2(2x-7)$

Apply product rule: $y\prime = f\prime g + fg\prime$.

$y' = (4(\sec(2x-1))^1\sec(2x-1)\tan(2x-1))(\cot(2x-7)) + ((\sec(2x-1))^2)(-2\csc^2(2x-7))$

$$\frac{dy}{dx} = 4(\sec(2x-1))^1\sec(2x-1)\tan(2x-1)\cot(2x-7) + (\sec(2x-1))^2 - 2\csc^2(2x-7)$$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

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S3

Differentiate with respect to x.

$y = \csc(8x + 5)\ \sin(-9x - 2)$

Answer:

$\frac{dy}{dx} = -8\ \csc(8x + 5)\cot(8x + 5)\ \sin(-9x - 2) + \csc(8x + 5)\ -9\ \cos(-9x - 2)$

Working:

Write the two factors as $f = \csc(8x + 5)$ and $g = \sin(-9x - 2)$.

Use product rule: $(fg)\prime = f\prime g + fg\prime$.

$f' = (-\csc(8x + 5)\cot(8x + 5))(8)$

$f' = -8\ \csc(8x + 5)\cot(8x + 5)$

$g' = (\cos(-9x - 2))(-9)$

$g' = -9\ \cos(-9x - 2)$

Substitute into the product rule: $y' = (-8\ \csc(8x + 5)\cot(8x + 5))(\sin(-9x - 2)) + (\csc(8x + 5))(-9\ \cos(-9x - 2))$

.

Final form: $\frac{dy}{dx} = -8\ \csc(8x + 5)\cot(8x + 5)\ \sin(-9x - 2) + \csc(8x + 5)\ -9\ \cos(-9x - 2)$

S4

Differentiate with respect to x.

$y = \frac{\sec(7x - 8)}{\sin(3x - 1)}$

Answer:

$\frac{dy}{dx} = \frac{(7\ \sec(7x - 8)\tan(7x - 8))\ \sin(3x - 1) - \sec(7x - 8)\ (3\ \cos(3x - 1))}{(\sin(3x - 1))^2}$

Working:

Identify the numerator and denominator: $u = \sec(7x - 8)$ and $v = \sin(3x - 1)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (\sec(7x - 8)\tan(7x - 8))(7)$

$u' = 7\ \sec(7x - 8)\tan(7x - 8)$

$v' = (\cos(3x - 1))(3)$

$v' = 3\ \cos(3x - 1)$

Substitute into the quotient rule before simplifying.

$\frac{dy}{dx} = \frac{(7\ \sec(7x - 8)\tan(7x - 8))\ \sin(3x - 1) - \sec(7x - 8)\ (3\ \cos(3x - 1))}{(\sin(3x - 1))^2}$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

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E1

Differentiate with respect to x.

$$y = \frac{\sin(-9x - 1)}{\sec(5x + 9)}$$

Answer:

$$\frac{dy}{dx} = \frac{(-9\cos(-9x - 1))\sec(5x + 9) - \sin(-9x - 1)(5\sec(5x + 9)\tan(5x + 9))}{(\sec(5x + 9))^2}$$

Working:

Identify the numerator and denominator: $u = \sin(-9x - 1)$ and $v = \sec(5x + 9)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (\cos(-9x - 1))(-9)$

$u' = -9\cos(-9x - 1)$

$v' = (\sec(5x + 9)\tan(5x + 9))(5)$

$v' = 5\sec(5x + 9)\tan(5x + 9)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(-9\cos(-9x - 1))\sec(5x + 9) - \sin(-9x - 1)(5\sec(5x + 9)\tan(5x + 9))}{(\sec(5x + 9))^2}$$

E2

Differentiate with respect to x.

$$y = (\sin(4x^2 - 5x - 2))^4$$

Answer:

$$\frac{dy}{dx} = 4(\sin(4x^2 - 5x - 2))^3\cos(4x^2 - 5x - 2)(8x - 5)$$

Working:

Work from the outside in: power, then trig function, then polynomial inner function.

Outer power derivative gives $4(\sin(4x^2 - 5x - 2))^3$.

Trig derivative gives $\cos(4x^2 - 5x - 2)$.

Inner derivative gives $8x - 5$.

Multiply the three derivative parts together.

$$\frac{dy}{dx} = 4(\sin(4x^2 - 5x - 2))^3\cos(4x^2 - 5x - 2)(8x - 5)$$

Teacher answers & working

Trigonometric Differentiation · Mixed Trig Challenges

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E3

Differentiate with respect to x.

$$y = \frac{\csc(6x+5)}{\sin(7x-2)}$$

Answer:

$$\frac{dy}{dx} = \frac{(-6\ \csc(6x+5)\cot(6x+5))\ \sin(7x-2) - \csc(6x+5)\ (7\ \cos(7x-2))}{(\sin(7x-2))^2}$$

Working:

Identify the numerator and denominator: $u = \csc(6x+5)$ and $v = \sin(7x-2)$.

Use quotient rule: $(u/v)\prime = (u\prime v - uv\prime)/v^2$.

$u' = (-\csc(6x+5)\cot(6x+5))(6)$

$u' = -6\ \csc(6x+5)\cot(6x+5)$

$v' = (\cos(7x-2))(7)$

$v' = 7\ \cos(7x-2)$

Substitute into the quotient rule before simplifying.

$$\frac{dy}{dx} = \frac{(-6\ \csc(6x+5)\cot(6x+5))\ \sin(7x-2) - \csc(6x+5)\ (7\ \cos(7x-2))}{(\sin(7x-2))^2}$$